Fraud Detection Report

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Technical Challenge

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Summary

The analysis of the 22,239 payment transactions (Nov 19-24, 2025) reveals a 77.75% approval rate with ≈€453K processed volume (in EUR).
Unsupervised machine learning identified critical behavioral patterns achieving high fraud detection precision, significantly outperforming traditional rule-based approaches (also resulting in overfitting at times).

Tools used:
- R (exploratory analysis, statistical testing)
- Python (One-Class SVM, DBSCAN/HDBSCAN, LOF, Isolation Forest; Random Forest+SMOTE as supervised benchmark)

Exploratory Data Analysis

Early assessment: while approval rate appears healthy, the 22.25% decline rate suggests overly conservative risk rules may be rejecting legitimate transactions alongside genuine threats.

Critical Limitation: only 6 days of data and 122 confirmed frauds is very limited data for the training of a robust and effective model.

Main Decline reasons

NOT_3D_AUTHENTICATED	1,301 transactions	€32.4K
BLOCKED BY MERCHANT RISK	884 transactions	€16.3K
DECLINED	797 transactions	€9.6K
NOT_ENOUGH_BALANCE	792 transactions	€10.0K
ISSUER_SUSPECTED_FRAUD	307 transactions	€8.4K

Pattern Analysis

<u>Transaction</u>

Fraud Concentration:
- 100% USD currency
- 97% transaction in the United States
- 86% VISA payment method
- Average amount: $42 (~1.8× the average non-fraudulent transaction)

Interpretation: organized card-testing rings targeting US-issued cards with repeated $10 and $250 top-ups. However, fraud patterns indicate exploitation of the authentication systems rather than bypassing them.

<u>Fraud</u>

Statistical testing revealed traditional fraud indicators fail to discriminate fraud cases: 99% of frauds passed 3D Secure authentication and CVC verification.

Interpretation: frauds are sophisticated, fraudsters successfully pass authentication checks. Individual and combined security features show weak effect sizes (Cramér's V≤0.15), pointing to the necessity of behavioral pattern analysis.

ML Fraud Detection

Methodology: unsupervised anomaly detection + supervised benchmark.

Model	Anomalies Detected	Fraud Recall (%)	Precision (%)	F1-Score (%)
DBSCAN	119	81.97	84.03	82.99
HDBSCAN	135	95.08	85.93	90.31
One-Class SVM	366	68.03	22.68	34.02
Isolation Forest Advanced	1,112	68.85	7.55	13.61
LOF	223	13.93	7.62	9.86
Random Forest + SMOTE	139	98.36	86.33	91.95

Table 1 - evaluation metrics for each model performance

Main Discoveries

Behavioral Signature

DBSCAN/HDBSCAN identified fraud clusters with high precision, flagging 19 suspicious transactions (pattern: BIN_frequency ~ 215, IP_frequency ~ 189, $10-$20 USD, US)
Interpretation: Automated card testing from fixed location. HDBSCAN detected 116/122 frauds (95%), incl. a second ring of $250 top-ups (both clustering methods converged on the same pattern)

Emerging Threat Detection

One-Class SVM detected 283 suspicious transactions worth ≈€8K (patterns: IP abuse, first-time cards); 68% fraud detection without historical labels points towards generalization to unknown threats.
Banking system is relatively efficient at detecting fraud, ISSUER_SUSPECTED_FRAUD flags 307 transactions, 53 of which present the clusters' behavioral signature pattern.
The merchant's system is less efficient: MERCHANT RISK RULES flag 884 transactions, 1 of which presents the clusters' behavioral pattern.